

Literature Status: Two Bishop-Property Questions

Run `fa_banach_001`, agent lane 13

2026-06-16

Status

literature_already_answered. Two open questions extracted from arXiv:1211.2770 are answered in the later paper arXiv:1310.4035. This packet records provenance only; it is not an original proof from the run.

Source Questions

The source paper is K. P. Hart, T. Kania, and T. Kochanek, *A chain condition for operators from $C(K)$ -spaces*, arXiv:1211.2770. In Section 3, Question 3.9 asks:

Does the class of compact Hausdorff spaces satisfying $(\mathfrak{B})$ contain all Eberlein compact spaces?

It then asks more generally whether every compact Hausdorff space K for which $C(K)$ is Lindelof in its weak topology satisfies $(\mathfrak{B})$.

In Section 4, Question 4.3 asks whether the set of operators on a $C(K)$ -space which satisfy $(\mathfrak{B})$ is always a right ideal, and hence a two-sided ideal, in $\mathcal{B}(C(K))$.

Supporting Answers

The supporting paper is T. Kania and R. J. Smith, *Chains of functions in $C(K)$ -spaces*, arXiv:1310.4035. Its abstract states that it answers two questions arising after $(\mathfrak{B})$ was introduced.

First, arXiv:1310.4035 proves Theorem 1.1: if a class $\mathcal{D}$ of compact Hausdorff spaces is preserved under closed subspaces and Hausdorff quotients and contains no non-metrizable linearly ordered space, then every member of $\mathcal{D}$ has $(\mathfrak{B})$. The introduction then applies this theorem to the class of compact spaces K for which $C_p(K)$ is Lindelof and explicitly says that this positively answers Question 3.9 from arXiv:1211.2770. Since all Eberlein compact spaces are Corson compact, and the paper recalls that Corson compact spaces lie in that $C_p(K)$ -Lindelof class, the Eberlein part is answered affirmatively.

Second, the final section of arXiv:1310.4035 explicitly returns to Question 4.3 of arXiv:1211.2770 and gives a negative example. With $B = B_{C(\beta\mathbb{N})^*}$, it constructs a compact space $B \sqcup \beta\mathbb{N}$ and operators S, T such that T has $(\mathfrak{B})$, while $TS^{-1} = I_{C(B \sqcup \beta\mathbb{N})}$ fails $(\mathfrak{B})$. Thus the Bishop operators need not form a right ideal.

Scope

The Eberlein question is answered positively. The more general weak-topology wording from arXiv:1211.2770 is not identical to the $C_p(K)$ -Lindelof class handled in arXiv:1310.4035, but the supporting paper explicitly records the positive answer to the Eberlein part and a broad pointwise-Lindelof strengthening. The right-ideal question is answered negatively.

This packet does not settle Question 2.6 of arXiv:1211.2770 about set algebras with the subsequential completeness property and vector measures satisfying the rook condition. A separate attempt note records that remaining hard core.

Search Evidence

The run's cheap indexes were searched for arXiv:1211.2770, the paper title, "Bishop", "right ideal", "Eberlein", "weakly Lindelof", "SCP", and "rook". No prior packet for arXiv:1211.2770 was found. Web search for the source title and the right-ideal/Eberlein phrases found arXiv:1310.4035, whose abstract and introduction explicitly identify the source questions it answers.

References

- [1] K. P. Hart, T. Kania, and T. Kochanek, *A chain condition for operators from $C(K)$ -spaces*, arXiv:1211.2770, 2012. <https://arxiv.org/abs/1211.2770>.
- [2] T. Kania and R. J. Smith, *Chains of functions in $C(K)$ -spaces*, arXiv:1310.4035, 2013. <https://arxiv.org/abs/1310.4035>.